

Principles of Applied Engineering

TEKS §127.781 • CTE Level 1 • Grade 9 • Mr. Gavaldon • Da Vinci School

Week 1

MONDAY

Day 1 - Procedures, Syllabus & Your Design Notebook

Aug 10

OBJECTIVE

Students will set up their engineering design notebook and explain how the class runs, then sign into Google Classroom.

TEKS

§127.781(d)(1) professional standards & employability; (d)(3)(B) maintain a design & computation engineering notebook

ELPS / EB SUPPORT

Model every notebook step live under the doc cam with icons + sentence stems (VC, CD, RS); pre-teach notebook/entry/table-of-contents (PV). A bilingual peer is available for sign-in as backup (PN) — the visual model carries it, not the peer.

AGENDA • 45 MIN

8' Welcome + seating + how the class runs

10' Syllabus + what we build this year

20' Set up your design notebook (cover, TOC, first entry)

7' Sign in to Google Classroom + exit ticket

SLIDES / ACTIVITY

Welcome to Engineering

NOTEBOOK

Set it up today: cover, Table of Contents (pgs 1-3), number pages, date entries. First entry = what you hope to build this year.

TURN IN

Design Notebook - Setup & Table of Contents

TUESDAY

Day 2 - What Is Engineering? (video + intro)

Aug 11

OBJECTIVE

Students will describe what engineers do, identify branches of engineering, and sketch/label one engineered object.

TEKS

§127.781(d)(2)(A) engineering disciplines & their history; (d)(2)(E) engineering career paths; (d)(3)(C) sketching to present ideas

ELPS / EB SUPPORT

Guided viewing sheet gives sentence stems (RS, CD). Pair-share before whole-class share so EB students rehearse first (PN, WT).

AGENDA • 45 MIN

5' Bell-work: list 3 engineered things you used today

5' Go over it

12' Watch: What is engineering? + guided notes

18' Branches mini-lesson + sketch & label task

5' Exit ticket

SLIDES / ACTIVITY

The World of Engineering

NOTEBOOK

Glue in the guided-viewing sheet. Sketch + label one engineered object; name its branch and the problem it solves.

TURN IN

Engineering All Around You - Guided Viewing + Sketch

WEDNESDAY

Day 3 - Paper Airplane Challenge (Design Process)

Aug 12

OBJECTIVE

Students will apply the engineering design process to build, test, and improve a paper airplane, recording each step.

TEKS

\$127.781(d)(2)(G) the engineering design process; (d)(1)(B) collaborate as a team

ELPS / EB SUPPORT

Design-process sheet is the scaffold (VC, CD). Mixed-level teams of 2-3 so EB students use the process vocabulary in context (PN).

AGENDA • 45 MIN

5' Bell-work: what makes a paper plane fly far?

10' Watch TED-Ed + name the 5 steps

8' Rules + brainstorm two designs

17' Build, test, improve (2 throws)

5' Record results + reflect

SLIDES / ACTIVITY

The Engineering Design Process

NOTEBOOK

Glue in the design-process record sheet. Sketch your two candidate designs before building.

TURN IN

Engineering Design Process - Record Sheet

THURSDAY

Day 4 - Paper Tower Challenge (iterate)

Aug 13

OBJECTIVE

Students will re-apply the design process to build the tallest free-standing paper tower and reflect on iteration.

TEKS

\$127.781(d)(2)(G) engineering design process — iteration & constraints; (d)(1)(B) teamwork

ELPS / EB SUPPORT

Same vocabulary reused (repetition builds language, RS). Roles (builder / recorder / tester) give every student a clear job (CD).

AGENDA • 45 MIN

5' Bell-work: which shape is strongest - and why?

5' Go over it + assign team roles

8' Constraints + plan

20' Build 1 -> improve -> Build 2 (measure both)

7' Reflection sheet + clean up

SLIDES / ACTIVITY

Build It Taller

NOTEBOOK

Log Build 1 and Build 2 heights. Note the one change you made and why. Sketch the strongest shape you found.

TURN IN

Tower Iteration - Reflection

FRIDAY

Day 5 - Types of Engineering + Quiz Game

Aug 14

OBJECTIVE

Students will match real products to the branch of engineering behind them and identify engineering around campus.

TEKS

\$127.781(d)(2)(A) fields of engineering; (d)(5) impact of technology on society

ELPS / EB SUPPORT

Team quiz lowers stakes; EB students answer with a peer (PN, WT). 'Spot the engineering' uses images, not text-heavy prompts (VC).

AGENDA • 45 MIN

5' Bell-work: match 3 products to a branch

10' Types of engineering mini-lesson

10' Team quiz game (match product -> branch)

15' Spot-the-engineering: campus walk OR in-room hunt (door, HVAC vent, outlets)

5' Share-out + exit

SLIDES / ACTIVITY

Engineering Is Everywhere

NOTEBOOK

Record your 3 campus finds: what it is, its branch, and the problem it solves.

TURN IN

Spot the Engineering - Campus Walk